Weak Quantum Decoding for Axon Tracts in the Tatikonda-Mitter Framework: a Proposal

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Abstract

In this note the authors examine the analysis of perception at the quantum level and provide a path to take it one step forward.

Consider the analysis of perception at the quantum level which was proposed in (Chawla 2024). In it we proposed that a metric perturbation g_1 impinging from one side onto an axon tract pair, would appear as g_2 in its effect on the second axon tract. Differing quantum information is thus transmitted along the two tracts. When this quantum information is received for decoding, again neural processes are involved. Thus the setup of weak information put forward in (Chawla and Morgera 2022) becomes relevant, but first has to be extended to the quantum setting (Wilde 2013). Further, in Section 9 of their paper on the capacity of channels with feedback (Tatikonda and Mitter 2008), Tatikonda and Mitter put forward maximum likelihood decoding in their abstract and rigorous probabilistic framework. Extending this to the quantum domain as a first step, and then setting up weak decoding in the new framework would be the appropriate way to handle the decoding of the two tracts' quantum information. This intriguing analysis will be taken up in future work.

References

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